CAS-Native AI Global Public Launch Timeline

Gated Roadmap From Research Prototype to Public LLM-Style Usage

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Gated Roadmap From Research Prototype to Public LLM-Style Usage

Starting point. April 29, 2026.

Purpose. This document gives stakeholders a practical launch roadmap for CAS-Native AI as a public LLM-style system. It explains the major intervals, what must be built, what must be proven, and what should not be claimed too early.

Important framing. This is a gated roadmap, not a guaranteed calendar promise. CAS-Native should move to wider release only when its validation, safety, infrastructure, and product-readiness gates clear.

Current status summary. CAS has a working compiler, a CAS Synthetic Data Generator, recoverable Level 3 prompt data, and an early CAS-Native reranker path. The strongest latest result is the patched recoverable reranker run: 21 / 30 exact top-1, 26 / 30 top-5, 2.89 mean rank, 28 / 30 slot-match, and 30 / 30 compiler-valid outputs on the current 30-item evaluation slice.

Executive Timeline

Phase	Approximate interval from Apr 29, 2026	Calendar window	Primary goal	Release posture
Phase 0	Current state	Apr 2026	Research prototype and evidence base	Internal only
Phase 1	0-30 days	May 2026	Evaluation hardening and meta_logic diagnosis	Internal / technical demos
Phase 2	1-3 months	Jun-Jul 2026	Closed alpha CAS-Native assistant	Invite-only research alpha
Phase 3	3-6 months	Aug-Oct 2026	Private beta with product workflows	Limited trusted users
Phase 4	6-9 months	Nov 2026-Jan 2027	Public research preview	Public but constrained
Phase 5	9-12 months	Feb-Apr 2027	Public beta API and app	Broader public usage
Phase 6	12-18 months	May-Oct 2027	Global public launch candidate	Production-scale public LLM-style usage
Phase 7	18-36 months	Nov 2027-Apr 2029	Multimodal CAS-native platform	Mature ecosystem

The aggressive path reaches a constrained public research preview in 6-9 months and a public beta in 9-12 months. A credible global public LLM-style launch should be treated as a 12-18 month target unless the validation program outperforms expectations.

What “Global Public Usage” Means

For stakeholders, “global public usage” should not mean merely putting a chatbot online.

It should mean:

1. **Reliable access.** Users can access CAS-Native through a web app and/or API with stable uptime.
2. **Compiler-grounded output.** CAS expressions are validated before being presented as formal outputs.
3. **Transparent uncertainty.** The system distinguishes exact recovery, likely interpretation, and ambiguous cases.
4. **Safe claims.** It does not present CAS interpretations as diagnosis, therapy, or behavioral certainty.
5. **Multimodal pathway.** Text, CAS notation, diagrams, voice, and visual artifacts can begin routing through a shared CAS representation layer.

6. **Auditability.** Outputs can be inspected, reproduced, scored, and traced to model, compiler, and candidate-selection behavior.

Phase 0: Current State

Where CAS-Native Is Today

Area	Current status
CAS compiler	Broad grammar, typed meta-logic, clusters, set operations, diagrams, Rule 4 validation
CAS Synthetic Data Generator	Level 2 generator plus Level 3 recoverable prompt pipeline
Recoverable dataset	19,454 compiler-valid Level 3 prompt-to-CAS examples
Semantic reranker	Patched run reached 21 / 30 exact top-1 and 26 / 30 top-5
Constrained inference	Candidate-bank and prompt-slot filtering path exists
Free-form CAS generation	Not product-ready; model can learn tokens but not reliably assemble grammar alone
Public CAS-Native LLM	Not launched yet

Main Lesson From Current Work

The right near-term path is not unconstrained generation.

The right path is:

```
natural-language prompt
  -> recoverable slot extraction
  -> compiler-valid candidate bank
  -> semantic reranker
  -> compiler validation
  -> CAS expression + explanation + diagram
```

This is closer to a structured reasoning engine than a standard chatbot.

Phase 1: Evaluation Hardening

Interval. 0-30 days from April 29, 2026. Target window: May 2026.

Goal. Turn the promising 30-item result into a defensible evaluation program.

Build Items

Workstream	What needs to happen	Why it matters
meta_logic diagnosis	Find why meta_logic lags at 2 / 5 exact and mean rank 15.67	It is the only major structural laggard
Larger held-out eval	Expand from 30 examples to 300-600 examples	30 examples is too small for serious claims
Paraphrase holdout	Test prompts in styles not seen during training	Detects shallow prompt-format overfitting
Candidate-bank audit	Measure target scoreability and filter attrition by bucket	Ensures correct answer is available to reranker
Error taxonomy	Label failures by candidate missing, filter error, semantic confusion, type ambiguity	Makes fixes targeted instead of speculative

Gates to Clear

Gate	Minimum target
Compiler-valid outputs	100% on constrained path
Exact top-1 on 300+ heldout examples	>= 60%
Ground-truth top-5 on 300+ heldout examples	>= 85%
Bucket with worst top-5	>= 70%
meta_logic target scoreability	>= 95%
Paraphrase holdout degradation	Less than 15 percentage points

Deliverable

A stakeholder-grade validation report:

```
CAS-Native Recoverable Prompt Evaluation v1
sample size: 300-600
by-bucket metrics
failure taxonomy
overfit checks
recommended release posture
```

Phase 2: Closed Alpha CAS-Native Assistant

Interval. 1-3 months from April 29, 2026. Target window: June-July 2026.

Goal. Build a closed alpha assistant that can translate, explain, and teach CAS under controlled conditions.

Core Product Capabilities

Capability	User-facing behavior	Technical basis
Natural language to CAS	User describes a structure; assistant returns CAS notation	Recoverable prompt pipeline + reranker
CAS to natural language	User enters expression; assistant explains it	Compiler AST + natural-language mirror templates
CAS diagrams	User gets visual structure diagrams	Compiler diagram payload + renderer
Invalid expression help	User sees why expression failed and how to fix it	Parser/typechecker error explanations
Tutorial mode	Guided learning path from basic states to meta-logic	Level 3 examples + glossary/dictionary

Alpha Restrictions

The alpha should not claim:

1. Clinical diagnosis.
2. Behavior prediction.
3. Autonomous therapy.
4. General human mind-reading.
5. Full frontier-model equivalence.

It should claim:

1. Formal CAS translation assistance.
2. Compiler-grounded notation validation.
3. Structured cognitive representation.
4. Early CAS-native learning and explanation.

Gates to Clear

Gate	Minimum target
Alpha user tasks completed without crash	>= 95%
CAS compiler validation for formal outputs	100% where formal output is claimed
User-rated explanation clarity	>= 4 / 5 in guided alpha
Ambiguous prompt handling	Clarifying question or multiple candidates, not false certainty
P0 safety issues	0 open

Phase 3: Private Beta With Product Workflows

Interval. 3-6 months from April 29, 2026. Target window: August-October 2026.

Goal. Move from a technical assistant to repeatable workflows for real users.

Product Workflows

Workflow	What the user does	CAS-Native output
Learn CAS	Works through guided examples	Explanations, diagrams, correction feedback
Analyze an idea	Enters a thought or project note	CAS interpretation + graph placement
Compare interpretations	Asks why one CAS expression differs from another	Side-by-side structural explanation
Build recipes	Describes outcome ingredients	Production expression + yield-style explanation
Explore clusters	Groups Noetics/states	Typed set/cluster diagrams

Infrastructure Work

Infrastructure area	Required work
Authentication	User accounts and access control
Logging	Privacy-aware output/error telemetry
Model routing	Separate compiler, reranker, and language explanation paths
Caching	Candidate-bank and compiler-result caching
Observability	Per-bucket accuracy and failure dashboards
Abuse controls	Rate limits and safety filters

Gates to Clear

Gate	Minimum target
Private beta users	50-200 trusted users
Formal output compiler-validity	100% on compiler-routed formal claims
Task success rate	>= 80%
Average latency for compiler/reranker path	< 5 seconds for common queries
Ambiguous prompt false-certainty incidents	< 2%

Phase 4: Public Research Preview

Interval. 6-9 months from April 29, 2026. Target window: November 2026-January 2027.

Goal. Public access with explicit research-preview limits.

What Ships

1. Public CAS compiler.
2. Natural-language-to-CAS assistant in constrained mode.
3. CAS-to-natural-language mirror explanations.
4. Diagram rendering.
5. Basic Idea Atlas integration.
6. Documentation, tutorials, glossary, and examples.
7. Public safety and limitations page.

What Does Not Ship Yet

1. Open-ended behavioral prediction.
2. Clinical/mental-health interpretation.
3. Unreviewed autonomous advice.
4. Unlimited API usage.
5. Claims of human-level psychological understanding.

Gates to Clear

Gate	Minimum target
Public uptime	>= 99% for preview window
Formal CAS outputs compiler-valid	100% where formal output is claimed
Heldout exact top-1	>= 70% on recoverable prompts
Heldout top-5	>= 90% on recoverable prompts
Human explanation quality	>= 4 / 5 average
Published limitation policy	Complete

Phase 5: Public Beta API and App

Interval. 9-12 months from April 29, 2026. Target window: February-April 2027.

Goal. Broader public usage with developer/API access and scaled infrastructure.

API Surface

Endpoint class	Function
Compile	Validate CAS expression, return AST, type, bytecode, diagram payload
Translate	Natural language to CAS candidates
Explain	CAS expression to natural-language explanation
Diagram	CAS expression to visual graph/containment/radial layout
Rerank	Rank compiler-valid candidates against a prompt
Tutorial	Return guided examples and feedback
Idea Atlas	Capture and map ideas

Production Infrastructure

Need	Requirement
Model serving	GPU or optimized CPU service for reranker/LLM paths
Compiler service	Low-latency deterministic parser/typechecker
Storage	User histories, idea graphs, audit logs
Privacy	Clear data retention and deletion policy
Versioning	CAS grammar version and model version attached to outputs
Monitoring	Accuracy drift, latency, error rates, safety events

Gates to Clear

Gate	Minimum target
API availability	>= 99.5%
Formal output validation	100% compiler-valid formal outputs
Public beta satisfaction	>= 4 / 5 from target users
Security review	Complete
Privacy review	Complete
Cost per query	Sustainable under pricing model

Phase 6: Global Public Launch Candidate

Interval. 12-18 months from April 29, 2026. Target window: May-October 2027.

Goal. Public LLM-style CAS-Native usage at meaningful scale.

Launch Candidate Capabilities

Capability	Expected launch-candidate behavior
Conversational CAS tutoring	Explains notation, corrects mistakes, teaches progressively
Natural-language CAS translation	Converts recoverable prompts into validated CAS expressions
CAS-native explanation	Maps every expression part to plain-language meaning
Diagram-native output	Shows graph, containment, radial, recipe, and dependency diagrams
Voice input	Speech-to-text routed into CAS interpretation
Voice output	Natural-language explanations read aloud
Vision input, limited	Images or diagrams can be described and mapped into CAS-relevant structures where appropriate
Multimodal output	Text, CAS notation, diagrams, and possibly sonification from shared CAS representation
API integration	Developers can build CAS-aware tools
Audit trail	Outputs include model/compiler/version metadata

Global Launch Gates

Gate	Minimum target
Formal output compiler-validity	100% for formal CAS claims
Recoverable prompt exact top-1	>= 80% on large heldout
Recoverable prompt top-5	>= 95% on large heldout
Worst-bucket top-5	>= 85%
Ambiguity handling	Clarifies or offers alternatives in >= 95% of ambiguous test cases
Production uptime	>= 99.9%
Security/privacy review	Passed
Public documentation	Complete
Support and incident process	Operational

Phase 7: Mature Multimodal CAS-Native Platform

Interval. 18-36 months from April 29, 2026. Target window: November 2027-April 2029.

Goal. CAS-Native becomes a platform where all modalities can route through shared CAS structure.

What "All Modalities Operating in All Modalities" Means

The long-term architecture should not treat text, image, voice, diagram, and audio as separate products.

It should treat CAS as the shared intermediate representation:

```
speech -> text -> CAS -> explanation
text -> CAS -> diagram
CAS -> natural language -> voice
image -> scene/diagram features -> CAS hypothesis
CAS -> radial diagram -> sonification
conversation history -> CAS memory graph -> future retrieval
```

Mature Capability Targets

Modality	Input	Output	CAS role
Text	Prompts, notes, dialogue	Explanations, summaries, CAS translations	Main interaction channel
Symbolic notation	CAS expressions	AST, validation, diagrams, explanations	Formal language channel
Diagram	User views or edits structure	Graph/containment/radial/recipe views	Visual reasoning channel
Voice	Spoken thoughts/questions	Spoken explanations	Accessibility and mobile capture
Image	Diagrams, sketches, screenshots, visual artifacts	CAS hypotheses and structured descriptions	Vision-to-structure bridge
Audio/sonification	Temporal Cognitive Score	Auditory representation	Pattern perception channel
Memory graph	Prior user ideas and CAS states	Retrieval, comparison, longitudinal maps	Continuity channel

Risk Register

Risk	Why it matters	Mitigation
Overfitting to recoverable prompt style	Model may perform well only on one template style	Paraphrase holdouts and external eval prompts
Candidate-bank gaps	Reranker cannot select missing ground truth	Scoreability audits and bank expansion
meta_logic lag	Advanced constructs may remain weak	Local diagnosis before more GPU training
Overclaiming	Stakeholder trust can be damaged	Use precise current/expected/hoped language
Latency	Public users will not tolerate slow responses	Cache compiler/candidates and optimize reranker service
Privacy	Idea Atlas and cognitive notes are sensitive	Data minimization, deletion, consent, encryption
Safety misuse	Users may treat CAS as diagnosis	Clear disclaimers and safety policy
Cost scaling	Public LLM usage can become expensive	Tiered model routing and compiler-first paths

Decision Points

Decision Point 1: After Phase 1

Question: Did the larger heldout evaluation confirm the patched reranker result?

If yes, proceed to closed alpha.

If no, do not launch. Diagnose data/candidate/filter failure.

Decision Point 2: After Phase 2

Question: Can real users complete guided CAS tasks without expert handholding?

If yes, proceed to private beta.

If no, improve tutorial, explanations, and error recovery.

Decision Point 3: After Phase 3

Question: Are workflows valuable outside the development team?

If yes, proceed to public research preview.

If no, narrow the product wedge.

Decision Point 4: Before Phase 6

Question: Is the system safe, useful, stable, and honest enough for global public usage?

If yes, launch candidate.

If no, keep it in public beta.

Recommended Stakeholder Framing

Use this short framing:

> CAS-Native AI is being built as a compiler-grounded cognitive language model. The launch plan is intentionally gated: first prove formal recovery and explanation, then closed alpha, then private beta, then public research preview, then global launch. The goal is not to release another generic chatbot. The goal is to release an AI system that can translate, validate, explain, diagram, and eventually reason across CAS structures in text, symbols, voice, vision, diagrams, and memory.

Practical Next Step

The immediate next step is not more public marketing.

The immediate next step is:

1. Terminate any idle GPU pod.
2. Diagnose the meta_logic bucket locally.
3. Patch candidate coverage or filtering if needed.
4. Run a larger heldout recoverable evaluation.
5. Only then move toward closed alpha.